



A07 · EXPLORING DEEP LEARNING
THROUGH POWERPOINT

Predicting HVAC Failures with a Neural Network

A simulated feed-forward network that learns whether a rooftop unit will fail within 30 days — and improves when it gets it wrong.

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ITAI 1370 · Prof. Devarakonda · Summer 2026 · Houston Community
College

The Problem — and the Network at a Glance

A supervised binary classifier: service-history features in, failure prediction out

1 THE PROBLEM

Commercial rooftop units (RTUs) fail without warning. Reactive repair means emergency labor rates, downtime, and lost product. Technicians already collect the data that predicts failure — it just isn't being used.

2 THE APPROACH

A feed-forward neural network trained on past service calls (supervised learning). Each historical record pairs unit features with a known outcome, so the network can learn the pattern.

3 THE ARCHITECTURE

4 input features → Hidden Layer 1 (4 neurons) → Hidden Layer 2 (3 neurons) → 2 output classes: Will Fail $\leq$ 30 Days / Will Not Fail.

THEORY · SUPERVISED LEARNING

The network learns a function mapping inputs to outputs from labeled examples — historical service calls where the outcome (failed / didn't fail) is already known. Depth (multiple hidden layers) lets it learn non-linear combinations of features that no single rule could capture.

Goodfellow, Bengio & Courville (2016), Deep Learning, ch. 5–6 · LeCun, Bengio & Hinton (2015), Nature 521

Input Layer

Four features the network analyzes — each one a signal a field technician already records

x1

Unit Age (yrs) Compressors and heat exchangers degrade with age; bearings and capacitors drift out of spec.

x2

Days Since Last Service Deferred maintenance is the #1 controllable failure driver — dirty coils and low charge compound.

x3

Avg Ambient Temp (°F) Houston heat: sustained high ambient pushes head pressure and duty cycle toward limits.

x4

Fault Codes (last 90 d) Board-logged faults (high-pressure trips, sensor errors) are early smoke before the fire.

THEORY · FEATURE SELECTION

Inputs are the network's entire view of the world — it can only learn patterns present in the features it is given.

These four were chosen for domain relevance (each has a known mechanical failure pathway) and availability (all are captured on routine service calls), following the principle that representation quality bounds model quality.

Goodfellow et al. (2016), ch. 5 — representation & features · LeCun et al. (2015)

The Dataset — Historical Service-Call Records

Five labeled examples (illustrative data · patterned on real commercial RTU service history)

Unit	Age (yrs)	Days Since Service	Avg Ambient (°F)	Fault Codes (90d)	Outcome (label)
RTU-01	12	210	98	4	FAILED
RTU-02	3	45	91	0	DID NOT FAIL
RTU-03	8	120	103	2	FAILED
RTU-04	15	30	95	1	DID NOT FAIL
RTU-05	6	300	99	3	FAILED

★ RTU-04 (highlighted) is the example we will push through the network — an old unit that was just serviced.

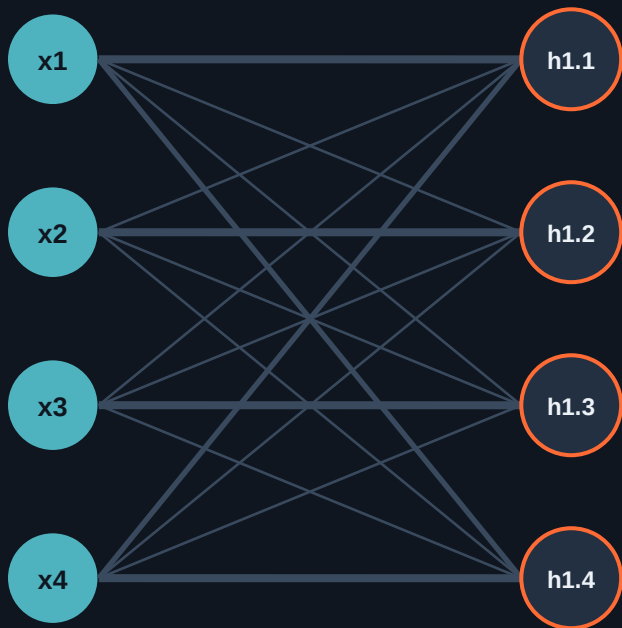
THEORY · TRAINING DATA & LABELS

Each row is a training example: a feature vector $x = [\text{age, recency, ambient, faults}]$ paired with a ground-truth label y . The network's job is to generalize from these pairs to unseen units — not memorize them.

Goodfellow et al. (2016), ch. 5 · LeCun et al. (2015), Nature 521

Hidden Layer 1

Four neurons — each learns a different combination of the raw inputs



INPUT LAYER

HIDDEN LAYER 1

THEORY · WEIGHTED SUMS

Each hidden neuron computes $z = \sum(w_i \cdot x_i) + b$ — a weighted vote over all inputs, plus a bias. Line thickness here represents $|w|$: thicker = more influence. Example: h1.1 weights unit age heavily; h1.3 specializes in ambient heat stress.

Goodfellow et al. (2016), ch. 6 — deep feedforward networks

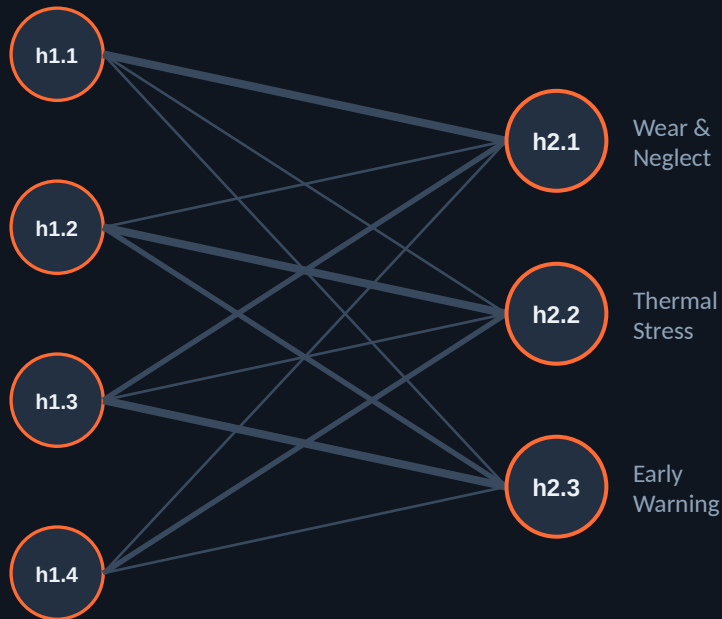
THEORY · ReLU ACTIVATION — a design decision

After the weighted sum, each neuron applies ReLU: $f(z) = \max(0, z)$. Chosen over sigmoid because ReLU avoids saturating gradients on positive inputs, keeping error signals strong during training in deeper stacks — critical for the learning step on slides 11–12.

Nair & Hinton (2010), ICML · Glorot, Bordes & Bengio (2011)

Hidden Layer 2

Three neurons — combining first-layer patterns into higher-level 'risk concepts'



HIDDEN LAYER 1

HIDDEN LAYER 2

THEORY · HIERARCHICAL REPRESENTATION

Depth is the point of deep learning: layer 1 detects simple feature combinations; layer 2 composes them into abstract concepts. Interpreting h2.1 as 'wear & neglect,' h2.2 as 'thermal stress,' and h2.3 as 'early warning signals' illustrates how networks build increasingly abstract representations layer by layer — the reason a 2-layer net can capture interactions (old AND neglected) that a single linear model cannot.

LeCun, Bengio & Hinton (2015), Nature 521, 436–444 · Goodfellow et al. (2016), ch. 6

Output Layer

Two prediction classes — the network's verdict on the unit



HIDDEN LAYER 2

OUTPUT LAYER

THEORY · OUTPUT & CONFIDENCE

The two output scores are converted to probabilities (softmax), e.g. $[0.78, 0.22]$ = 78% confidence in failure.

The dispatch decision then becomes a threshold call: schedule preventive service when $P(\text{fail})$ crosses a level the business chooses — trading false alarms against missed failures.

Goodfellow et al. (2016), ch. 6.2 — output units & softmax

Activate the Input Layer — Feeding RTU-04 In

Highlighting the data that matches our chosen example from the dataset

x1 Unit Age

15 yrs

VERY HIGH — pushes toward failure

x2 Days Since Service

30 days

LOW — freshly maintained, protective

x3 Avg Ambient Temp

95 °F

ELEVATED — Houston summer load

x4 Fault Codes (90d)

1

LOW — minor signal only

THEORY · INPUT ENCODING

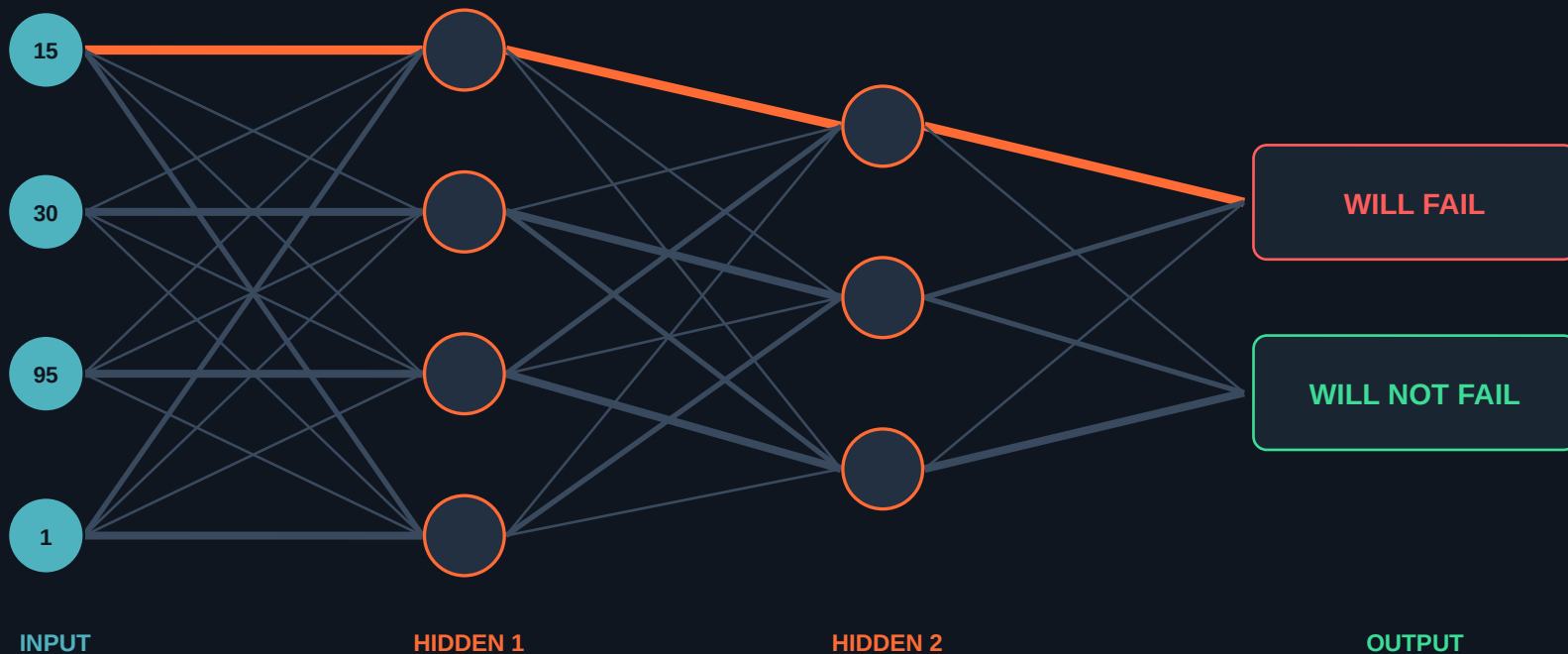
Raw field values are normalized before entering the network (e.g., scaled 0–1) so no feature dominates purely by its unit of measure.

Notice the tension in this example: age screams risk while service recency says safe. What the network decides reveals which connections it currently trusts — and sets up the learning moment.

Goodfellow et al. (2016), ch. 8.7.1 — input normalization

Processing Through the Layers

Line thickness = connection weight $|w|$ — watch where RTU-04's signal flows strongest

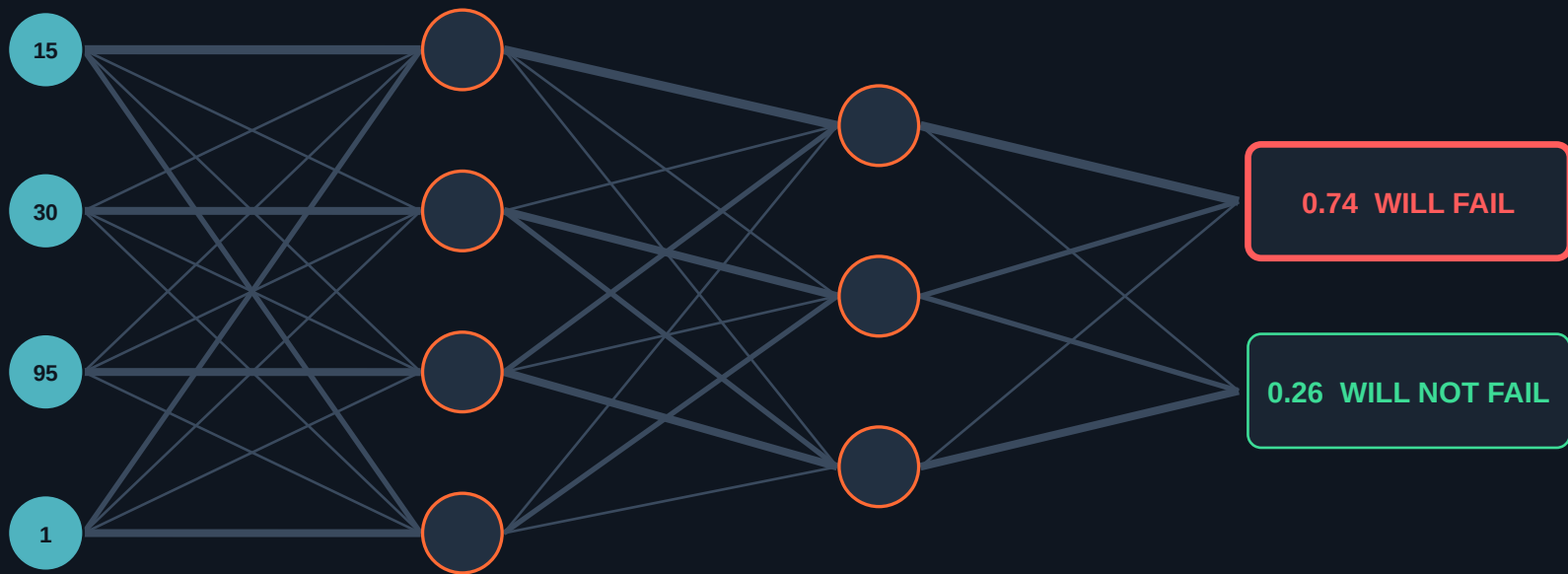


THEORY · FORWARD PROPAGATION

Each layer computes $\text{ReLU}(W \cdot a + b)$ on the previous layer's output. Orange = the dominant route for this input: $\text{age}(15) \rightarrow h1.1 \rightarrow h2.1 \rightarrow \text{WILL FAIL}$, since age carries the largest weights. · Goodfellow et al. (2016), ch. 6.5

The Network's Prediction

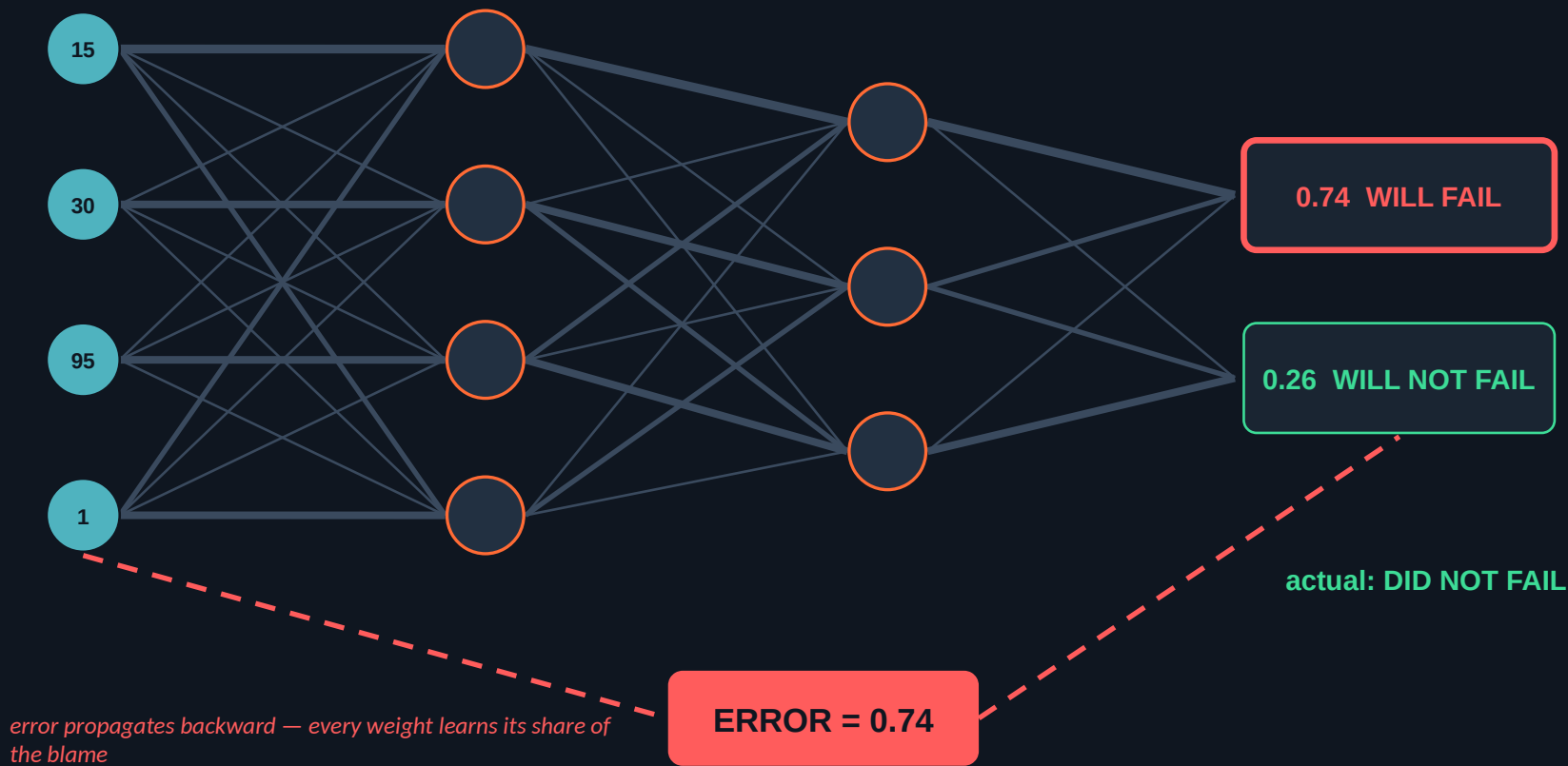
Softmax over the output scores — the age-heavy weighting drives the verdict



PREDICTION: WILL FAIL WITHIN 30 DAYS (74%) — driven almost entirely by Unit Age = 15 through the h1.1 → h2.1 'wear & neglect' path

Feedback Loop — Reality Checks the Network

RTU-04 did NOT fail. The prediction was wrong — and that error is the fuel for learning



Learning From the Error — Backprop Adjusts Weights

How theory says the connections should change after the RTU-04 miss



Age → h1.1 (was the loudest voice) · WEAKENED

Gradient assigns this connection the largest share of blame: it drove a confident wrong answer. Its weight shrinks proportionally.



Days-Since-Service → h1.2 · STRENGTHENED

Recent service was the true protective signal the network under-weighted. Its gradient pushes the weight up.



h2.1 'Wear & Neglect' → WILL FAIL · MODERATED

The concept neuron stays useful, but its vote toward failure is tempered so age alone can no longer dominate.

THEORY · BACKPROP + GRADIENT DESCENT

Backpropagation uses the chain rule to compute $\partial \text{Error} / \partial w$ for every weight — how much each connection contributed to the miss. Gradient descent then updates each: $w \leftarrow w - \eta \cdot \partial E / \partial w$.

The learning rate η controls step size: too large overshoots, too small crawls. One example nudges weights; over many examples, the pattern converges.

Rumelhart, Hinton & Williams (1986), Nature 323 · Goodfellow et al. (2016), ch. 6.5, 8

AFTER LEARNING: re-running RTU-04 now yields $P(\text{fail}) \approx 0.41 \rightarrow$ prediction flips to WILL NOT FAIL. The network learned that a freshly-serviced old unit $\neq$ a neglected old unit — an interaction no single weight could express before.

Discussion Points

The three questions the assignment asks — answered

Q1

How theory set the connection strengths

Initial weights encoded domain priors (age matters), then deep learning theory took over: weighted sums (Goodfellow ch. 6) defined influence, ReLU (Nair & Hinton 2010) kept gradients alive, and every strengthen/weaken decision after the miss followed gradient attribution — not intuition.

Q2

What the error taught us

A 0.74-confident miss revealed over-reliance on one feature. Backpropagation (Rumelhart et al. 1986) converted that single error into targeted, proportional weight updates. Theoretical guides for further accuracy: tune learning rate η , train on more labeled service calls, and hold out a validation set to detect overfitting.

Q3

Real-world applications of this exact pattern

Predictive maintenance on commercial RTUs and chillers · cleanroom environmental-monitoring anomaly detection (pharma/biotech HVAC) · smart-building BAS optimization · dispatch prioritization from live fault-code telemetry — the same architecture, larger datasets.

References

Theoretical foundations cited throughout

Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press. <https://www.deeplearningbook.org> — feedforward networks (ch. 6), training & learning rate (ch. 8), supervised learning (ch. 5).

Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, 323, 533–536. — the backpropagation algorithm.

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521, 436–444. — hierarchical representation learning across layers.

Nair, V., & Hinton, G. E. (2010). Rectified linear units improve restricted Boltzmann machines. *Proceedings of ICML-10*. — ReLU activation choice.

Glorot, X., Bordes, A., & Bengio, Y. (2011). Deep sparse rectifier neural networks. *Proceedings of AISTATS*. — rectifier activations in deep stacks.

Dataset: illustrative service-call records patterned on commercial RTU field experience (TuringCollective). Deck built with an AI-assisted pipeline (documented in CLAUDE.md) per course AI policy.